

CoopIS 2026 - Paper #52

Paper Title: *Task Recognition from Physiological Data Using Multimodal Sensor Fusion For Human-Robot Collaboration*

Reviewer #1

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 4
Originality (Newness of the ideas expressed): 3
Technical Quality (Theoretical soundness/methodology): 1
Significance (Is the problem worth the given attention?): 1
Presentation (Structure/Length/English): 1
Overall Rating (Weighted value of above items): 2

Reviewer (self)

Compliance with INSTICC policy on usage of Generative AI in Reviews Yes
Reviewer degree of specialization in the area Average

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes
Needs more experimental results? Yes
Needs comparative evaluation? Yes
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): No
Conclusions/Future Work are convincing? Yes
References are up-to-date and appropriate? No
Paper formatting needs adjustment? No
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The paper investigates the identification of task types to support human–robot collaboration.

The topic is interesting and relevant with the increase development and use of robots in the field .
Unfortunately there are some shortcomings that need to be addressed:

- It is not clear how task types have been identified . There is no discussion about which task–types and how they have been identified. I assume they have been labelled already in the dataset but it is not discussed or described.
- It is important to describe how this approach can be applied in the real world and what kind of impact it can have.
- There are many formulas included but they often not sufficiently described. It is important to describe all variables and motivate them.

- The results cannot be replicated because no code seem to be provided.
- There is no discussion about the significance of the numbers.
- The related work refers to "recent" work which is 7, 5 and 2 years old.

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Reviewer #2

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 5
Originality (Newness of the ideas expressed): 2
Technical Quality (Theoretical soundness/methodology): 2
Significance (Is the problem worth the given attention?): 5
Presentation (Structure/Length/English): 1
Overall Rating (Weighted value of above items): 2

Reviewer (self)

Compliance with INSTICC policy on usage of Generative AI in Reviews Yes
Reviewer degree of specialization in the area Low

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Too short
Needs more experimental results? No
Needs comparative evaluation? No
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): Yes
Conclusions/Future Work are convincing? No
References are up-to-date and appropriate? Yes
Paper formatting needs adjustment? No
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The paper addresses task recognition in Human–Robot Collaboration using multimodal physiological data. The proposed approach combines preprocessing, multimodal feature fusion, and different classification models, with XGBoost obtaining the best reported performance.

The paper addresses an interesting problem in Human–Robot Collaboration, namely task recognition from multimodal physiological data. However, the main contribution of the position paper is not sufficiently clear.

The authors state that they propose a "compact multimodal fusion pipeline", but it is difficult to understand what is actually new in this pipeline and how it differs from the application and combination of existing preprocessing, feature fusion, and classification techniques. Since this is a position paper, the proposed research idea and its novelty should be stated more explicitly.

More details would also help in understanding how the proposed pipeline works, particularly the multimodal fusion step and the relationship between the different components. While a complete implementation and extensive validation may not be expected at this stage, the approach should be described in enough detail to assess its soundness and potential contribution.

Finally, the organization of the evaluation is confusing. Section 3.6 and Section 5 are both titled "Evaluation", although they appear to serve different purposes. Their roles should be better distinguished.

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Reviewer #3

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 4
Originality (Newness of the ideas expressed): 2
Technical Quality (Theoretical soundness/methodology): 2
Significance (Is the problem worth the given attention?): 2
Presentation (Structure/Length/English): 3
Overall Rating (Weighted value of above items): 2

Reviewer (self)

Compliance with INSTICC policy on usage of Generative AI in Reviews Yes
Reviewer degree of specialization in the area High

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? No
Needs more experimental results? Yes
Needs comparative evaluation? Yes
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): No
Conclusions/Future Work are convincing? No
References are up-to-date and appropriate? No
Paper formatting needs adjustment? Yes
Improve English? Yes

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

This paper studies five-class task recognition from multimodal physiological data for adaptive human-robot collaboration. It compares several established machine-learning models under a common stated preprocessing and hold-out protocol. The use of a recent public dataset, train-only preprocessing, imbalance-aware metrics, and class-level diagnostics provides a useful starting point. XGBoost reports the strongest results, with 0.9371 accuracy and 0.9142 macro-F1, and the displayed aggregate and per-class arithmetic is internally coherent. The paper is also concise and generally well presented.

The central claims would benefit from stronger substantiation. Reference [2], Bussolan et al., "MultiPhysio-HRC," Robotics 2025, 14(12), 184 evaluates stress and cognitive load as separate Low, Medium, and High tasks using Leave-One-Subject-Out evaluation, with published F1 ranges of 0.34 to 0.39 and 0.33 to 0.49, respectively. This paper instead reports a five-class schema without a reproducible mapping or alignment

procedure, so its 0.9142 macro-F1 is not directly comparable. The stratified 80/20 split also does not identify participant, session, trial, or window grouping. A row-level split of repeated observations would create leakage risk, although overlap or inflation is not proven. Participant-disjoint evaluation with split identifiers and uncertainty would better support the across-operator claim. Single-modality and no-fusion ablations are needed to demonstrate fusion benefit. Because the study is offline, the HRC claims could be narrowed or supported with latency, adaptation, safety, fallback, and robot-interface evidence.

Reproducibility could be improved by reporting participants, sessions, exclusions, sample and class counts, window construction, synchronization, feature blocks, missingness, class weights, model hyperparameters, seed, tuning procedure, and software configuration. Raw confusion counts, repeated-run intervals, calibration, paired comparisons, simple baselines, and evaluation under the source protocol would make the results easier to interpret. Figure 1 should also be reconciled with the models described in the text, particularly its sequence-encoding and BiLSTM path. The roadmap, duplicate Evaluation heading, incomplete abstract label list, repeated pipeline wording, and inaccurate reference records should be corrected. A focused threats-to-validity discussion and clearer treatment of privacy and workplace use would further strengthen the paper.

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Reviewer #4

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 2
Originality (Newness of the ideas expressed): 2
Technical Quality (Theoretical soundness/methodology): 2
Significance (Is the problem worth the given attention?): 3
Presentation (Structure/Length/English): 4
Overall Rating (Weighted value of above items): 2

Reviewer (self)

Compliance with INSTICC policy on usage of Generative AI in Reviews Yes
Reviewer degree of specialization in the area Average

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes
Needs more experimental results? Yes
Needs comparative evaluation? Yes
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): Yes
Conclusions/Future Work are convincing? No
References are up-to-date and appropriate? No
Paper formatting needs adjustment? No
Improve English? No

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The paper discusses Human Robotic Collaborations. In these it would be beneficial to detect certain states that the operator is in, i.e., cognitive load, high stress, Operational or work-oriented task, low load, ... The authors propose to collect this data from sensors such as EEG, EMG, EDA, ECG, respiration rate, ... to finally classify the task types.

The paper could be more self-contained. "Industrial task" is the one that is hard to understand. [2] doesn't help much. It could be related to manual labor, whereas all the other task types have to do with the operators physiological/cognitive state?!

How is MultiPhysio-HRC mapped to the 5 task-types utilized in this paper?

The paper promises "differences between operators, changes in sensor relationships over time, and imbalanced label distributions." Eq.1 addresses the last, the rest is missing.

[3] (Collot et al., "Balanced accuracy: The right metric for evaluating LLM judges") What? This is an LLM-as-a-judge paper? How does this explain balanced accuracy in your paper? This needs much more explanation. While we are at it: why are [4] and [5] in the paper. They are about continual learning and catastrophic forgetting. What is the significance for this paper?

The fused multimodal input should be evaluated against single-modalities baseline (e.g., EDA-only or ECG-only). How else could the authors justify that the fusion actually helps?

Also in table 2 MLP has 0.8727 for both Balanced Acc. and Macro-F1. Could this be a copy and paste error (two different aggregate statistics yielding the same value seems implausible).

This work can not be reproduced. There is no code, or data. We do not get hyperparameters for any of the five models, and no feature dimensionality d is state anywhere.